

E-Commerce Data Analysis | Portfolio Project

Problem Statement

Analyzed 99K+ orders from a Brazilian e-commerce platform to uncover revenue drivers, customer behavior patterns, product performance gaps, and delivery inefficiencies — enabling data-driven business decisions.

Tools Used

MySQL · Python (Pandas, Seaborn) · Power BI

What I Did

- Designed a **7-table relational database schema** and wrote 40+ SQL queries covering joins, aggregations, window functions, CTEs, and date functions
- Performed **EDA in Python** to validate data quality and visualize distributions
- Built a **4-page interactive Power BI dashboard** with KPI cards, trend charts, pie charts, and bar charts

Key Findings

Finding	Impact
97.02% order success rate	Strong logistics performance
health_beauty = top category (₹1.26M)	Prioritize inventory & promotions
Credit card used by 73.92% customers	Optimize checkout for card payments
Only 3.12% repeat customers	Loyalty program needed urgently
São Paulo = 42K customers	Focus marketing spend on SP

Results Snapshot

- ₹16M** total revenue analyzed
- 96K** unique customers profiled
- 4.09/5** average product rating
- 113K** products sold across categories

Skills Demonstrated

SQL Joins Window Functions Date Functions Data Modeling EDA Dashboard Design
Business Insights KPI Tracking Storytelling with Data